



RESEARCH ARTICLE | NOVEMBER 05 2024

Electron temperature and ion density distribution on a vertical section in a weakly magnetized inductively coupled plasma

You He  ; Yi-Lang Jiang  ; Myoung-Jae Lee  ; Beom-Jun Seo  ; Ju-Ho Kim   ; Chin-Wook Chung  



J. Vac. Sci. Technol. A 42, 063004 (2024)

<https://doi.org/10.1116/6.0003874>



Articles You May Be Interested In

Effect of a DC gradient magnetic field on electron density in a weakly magnetized inductively coupled plasma

J. Vac. Sci. Technol. A (October 2024)

Highly efficient plasma generation in inductively coupled plasmas using a parallel capacitor

J. Vac. Sci. Technol. A (December 2022)

Self-consistent simulation of electron cyclotron resonance plasma discharge

Phys. Plasmas (July 2000)

15 September 2025 21:32:22

Instruments for Advanced Science

HIDEN
ANALYTICAL

- Knowledge
- Experience
- Expertise

Click to view our product catalogue

Contact Hiden Analytical for further details:
 www.HidenAnalytical.com
 info@hiden.co.uk



Gas Analysis

- dynamic measurement of reaction gas streams
- catalysis and thermal analysis
- molecular beam studies
- dissolved species probes
- fermentation, environmental and ecological studies



Surface Science

- UHV/TPD
- SIMS
- end point detection in ion beam etch
- elemental imaging - surface mapping



Plasma Diagnostics

- plasma source characterization
- etch and deposition process reaction kinetic studies
- analysis of neutral and radical species



Vacuum Analysis

- partial pressure measurement and control of process gases
- reactive sputter process control
- vacuum diagnostics
- vacuum coating process monitoring

Electron temperature and ion density distribution on a vertical section in a weakly magnetized inductively coupled plasma

Cite as: J. Vac. Sci. Technol. A 42, 063004 (2024); doi: [10.1116/6.0003874](https://doi.org/10.1116/6.0003874)

Submitted: 3 July 2024 · Accepted: 14 October 2024 ·

Published Online: 5 November 2024



You He,¹ Yi-Lang Jiang,¹ Myoung-Jae Lee,² Beom-Jun Seo,¹ Ju-Ho Kim,^{1,a)}
and Chin-Wook Chung^{1,a)}

AFFILIATIONS

¹Department of Electrical Engineering, Hanyang University, 17 Haengdang-dong, Seongdong-gu, Seoul 133-791, Republic of Korea

²Department of Physics, Hanyang University, 17 Haengdang-dong, Seongdong-gu, Seoul 133-791, Republic of Korea

^{a)}Authors to whom correspondence should be addressed: electrickjh@gmail.com and joykang@hanyang.ac.kr

ABSTRACT

In this study, the distributions of electron temperature and ion density on a vertical section in a weakly magnetized inductively coupled plasma were measured using radially movable floating probes placed at different axial positions. The chamber used in this experiment included two cylindrical parts: a smaller radius top part with a planar antenna on the top quartz window and a larger radius downstream part. A magnet coil around the chamber top part maintained a divergent magnetic field in the discharge region. As the current in the magnet coil increased, the magnetic field also increased. Due to the variations of the radio frequency electric field in the plasma, the increase in electron temperature can be divided into different stages. At the higher magnetic field, the electric field of the electrostatic wave can increase electron temperature at the chamber center axial. Also, since the electron cyclotron resonance (ECR) heating in the chamber downstream part changed with the magnetic field, the maximum ion density was observed when the magnetic field around the bias electrode was slightly larger than the ECR magnetic condition. The reasons for these variations were verified in the plasma numerical simulations. The ion flux distribution measured on the bias electrode can change from a center-high distribution to an M-shape distribution with the increased magnetic field.

Published under an exclusive license by the AVS. <https://doi.org/10.1116/6.0003874>

I. INTRODUCTION

Weakly magnetized inductively coupled plasma (ICP) is used for plasma processing in semiconductor manufacturing.^{1,2} Applying a DC magnetic field in an ICP can increase the electron (or ion) density due to the confinement effect and can also increase the electron temperature by electron cyclotron resonance (ECR) heating.^{3,4} Also, the higher electron density and electron temperature can improve plasma processing.^{5–8} Compared with microwave ECR plasma sources, the required magnetic field can be reduced when the driving frequency decreases to radio frequency (RF).⁹ The weaker magnetic field of the RF-ECR source can significantly reduce the degree of anisotropy of the electron energy distribution.^{9,10} Homogenous and inhomogeneous DC magnetic fields can be generated by adjusting the position, size, and current of the magnet coils. Weakly magnetized ICPs in a homogeneous DC magnetic field have been studied.^{6,11,12}

An ECR chamber generally consists of a top cylindrical part with a smaller radius as a source region and another downstream cylindrical part with a larger radius as a diffusion region to place the wafer on a bias electrode.^{13–18} The antenna and the magnet coil are located around the chamber top part, and a divergent DC magnetic field is achieved from the antenna to the bias electrode.^{19–21} Within the chamber, this divergent DC magnetic field means that the magnitude, direction, and gradient of the DC magnetic field are different at each position. These lead to the differences in the RF electric field, power deposition, and plasma diffusion. Ultimately, the distributions of electron temperature and ion density can change with the current in the magnet coil. Therefore, it is necessary to measure the distributions of electron temperature and ion density on a vertical section in an ICP with a divergent DC magnetic field and to analyze the variation of these distributions versus the magnet coil current. However, previous studies have focused on

15 September 2025 21:32:22

the radial distribution in the chamber downstream (diffusion) part^{22–25} or the chamber top (source) part²⁶ and the axial distribution on the central axis of the chamber.^{22,23} The distributions of electron temperature, electron density, and plasma potential on a vertical section were measured in an ICP with a divergent magnetic field created by the permanent magnets.²⁷

In this work, the distributions of electron temperature and ion density on a vertical section in an ICP with a divergent DC magnetic field were measured using floating probes. The chamber used was composed of a top cylindrical part with a smaller radius and a downstream cylindrical part with a larger radius, and a magnet coil was placed around the chamber top part. The floating probes were placed at different axial positions and could be moved radially in the chamber. A numerical model of the magnetized ICP considering ECR heating was calculated to analyze the variations of the distributions. With the increase of the magnet coil current, the magnetic field in the chamber also increased. Due to the variation of the RF electric field, the electron temperature below the antenna increased first, and then a higher electron temperature was observed in the entire chamber at a larger magnet coil current. The electric field of the electrostatic wave can increase electron temperature at the chamber center axial at the higher magnetic field. Because the absorbed power by ECR heating in the chamber downstream part changed with the magnetic field, ion density increased first, the maximum ion density was observed when the magnetic field around the bias electrode was slightly larger than the ECR magnetic condition, and the ion density decreased at the larger magnetic field. On the bias electrode, the measured ion flux changed as a function of the electron temperature and ion density, and the ion flux distribution can change from a center-high distribution to an M-shape distribution with the increased magnetic field.

II. EXPERIMENTAL SETUP

The experimental procedures and calculations must be described clearly and sufficient information should be given such that they could be reproduced by other scientists.

As shown in Fig. 1, the chamber consists of two cylindrical parts. The top part of the chamber, as the source region, was made of quartz with an 85 mm radius and 60 mm height. A two-turn planar antenna was connected to a matching network and a 60 MHz generator (Seren HR1001) operating at 300 W of RF power. For the downstream part of the chamber, the radius was 185 mm, and the gap length between the top plate and the 2D probe on the bias electrode was 60 mm. The top plate and the side wall of the chamber downstream part were grounded. Argon gas was injected into the chamber at a mass flow of 10 SCCM, and the pressure was maintained at 5 mTorr through a pumping system.

The center of the 2D probe surface was regarded as the origin. The upward direction of the chamber, marked by the red arrow in Fig. 1(a), was considered the positive axial direction. Three planar probe tips with 6 mm of diameter were set to the axial positions (z) of 22, 45, and 85 mm, respectively. These three tips could be moved in the radial direction to measure the electron temperature (T_e) and ion density (n_i) at different radial positions (r). At z of 85 mm, the measurement was performed at r between 0 and 75 mm with an interval of 15 mm; at z values of 45 and 22 mm, the probe tips moved from 0 to 150 mm of r with an interval of 30 mm in this experiment. The wafer-type 2D probe²⁸ was installed on the bias electrode, and the planar probe tip array is shown in Fig. 1(b). The probe tips with 10 mm of diameter were installed on the bias electrode in the range of -135 to 135 mm of r . The surface of the 2D probe, except for the planar probe tips, was electrically floated due to the dielectric coating on it. Considering that the distributions of T_e and n_i in a cylindrical chamber are approximately axially symmetric, the distributions of T_e and n_i on the vertical section of the chamber could be observed by the probes in the range of r from -150 to 150 mm and of z from 0 to 85 mm.

T_e and n_i were measured by the Langmuir probe based on the floating harmonic method.²⁹ Two RF choke filters with high impedance at 60 and 120 MHz were connected in series to the probe tip to compensate for the RF voltage fluctuations. Due to the DC blocking capacitor in the floating probes,²⁹ the potential of the probe tip was maintained at floating potential. When a small

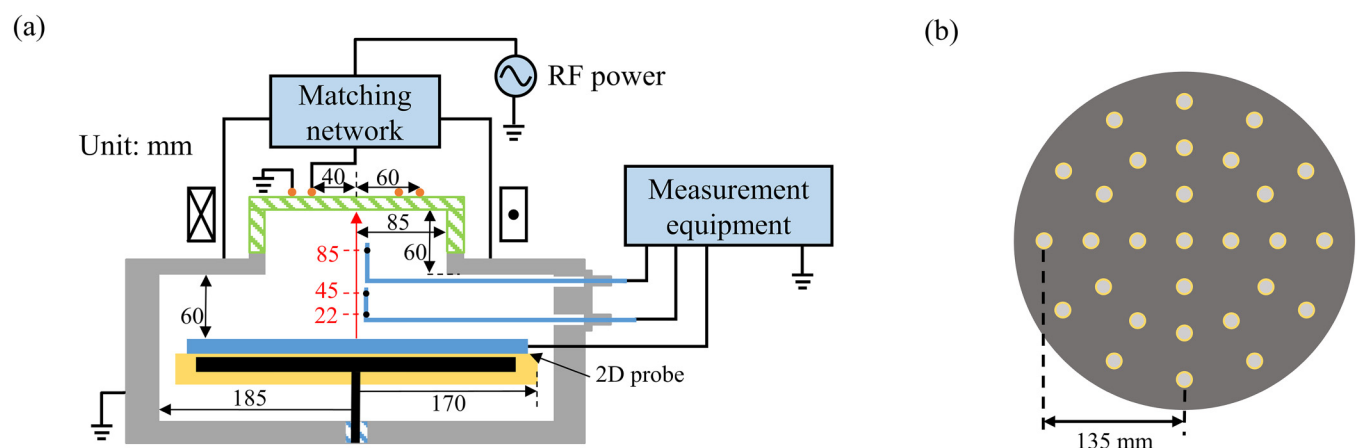


FIG. 1. (a) Experimental setup. (b) Planar tip array of the 2D probe.

15 September 2025 21:32:22

sinusoidal voltage with an amplitude of 1 V and a frequency of 10 kHz ($v_0 \cos(\omega t)$) was applied to the floating probe, due to the nonlinearity of the sheath, the probe current could be expanded by many harmonic terms. The amplitudes of the first and second harmonic currents (I_{ω} and $I_{2\omega}$) can be used to calculate T_e as follows:²⁹

$$\frac{I_{\omega}}{I_{2\omega}} = \frac{I_1(v_0/T_e)}{I_2(v_0/T_e)} \rightarrow T_e. \quad (1)$$

Here, I_x is the x -order-modified Bessel function. n_i can be calculated from the measured first harmonic current as follows:²⁹

$$n_i = \frac{I_{\omega}}{2(0.61eu_B A_p)} \frac{I_0(v_0/T_e)}{I_1(v_0/T_e)}. \quad (2)$$

Here, e is the unit charge, u_B is the Bohm velocity, and A_p is the area of the planar probe tip. Due to the potential drop across a collisionless presheath, the ratio of n_i at the sheath edge to the bulk plasma is approximately 0.61.

A magnet coil was used to maintain the DC magnetic field (B) in the chamber. Because the central axis of the magnet coil coincided with the central axis of the chamber, the component of the magnetic field in the radial direction was 0 G, and the magnetic field (B) on the central axis was measured and shown by the symbols in Fig. 2. As the current applied in the magnet coil (I_{mag}) increased, B almost linearly increased for each z . B was also calculated in a numerical model with the same geometrics and I_{mag} as the experiment, and the results are shown by the lines in Fig. 2. The magnitudes and the trends of B versus I_{mag} were similar between the simulation and the measurement. However, at $z = 0$ mm, a difference in B between the simulation and the measurement was observed because the arrangement of the wires in the magnet coil was not considered in the simulation. Because the

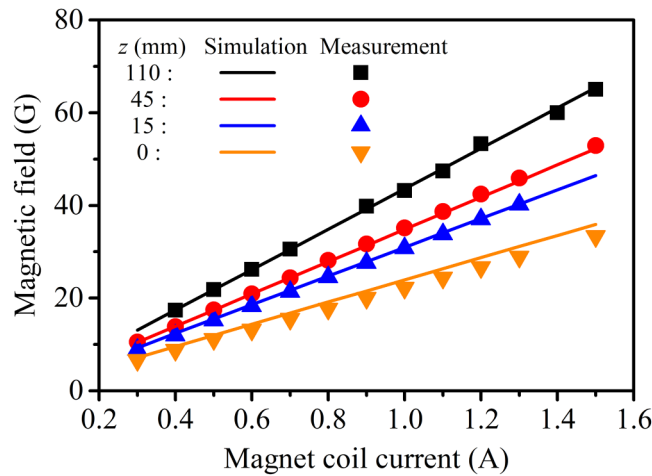


FIG. 2. Magnetic field vs magnet coil current at different positions on the chamber central axis. The lines and symbols were obtained by simulation and measurement.

radius of the magnet coil was smaller than that of the chamber downstream part, B in the discharge region of the chamber diverged from the top part to the downstream part. The magnitude and the direction of B on the vertical section of the chamber for I_{mag} values of 0.4, 0.6, 0.8, and 1.5 A were calculated and are shown in Figs. 3(a)–3(d), respectively. In Fig. 3, the color map indicates the magnitude of B , the arrows indicate the direction of B , the black line marks the edges of the chamber and the magnet coil, and $B = 21.4$ G (which is the ECR magnetic condition at 60 MHz of driving frequency) is highlighted by the white dotted line for each I_{mag} . As I_{mag} increased, the magnitude of B increased, while the direction of B did not change. When the 0.4 A of I_{mag} was applied, B was less than 21.4 G for almost the entire chamber. The position where B was 21.4 G shifted from the chamber top part to the bias electrode with increasing I_{mag} , and B was larger than 21.4 G in the entire chamber at $I_{\text{mag}} = 1.5$ A. B in the experiment and the simulation of this research was in the range of 0.4 to 1.5 A.

III. SIMULATION SETUP

A numerical simulation of the magnetized ICP was performed in COMSOL Multiphysics. A two-dimensional (2D) model with the same geometry as the chamber used in the experiment was simulated. B in the simulation was set to that measured in the experiment. The RF electric field ($\vec{E}$) in the simulation was calculated by using the wave equation considering the plasma dielectric constant tensor, $\hat{\epsilon}_p$, and the permeability in the vacuum, μ_0 ,

$$\nabla^2 \times \vec{E} + \omega_{\text{RF}}^2 \mu_0 \hat{\epsilon}_p \vec{E} = 0. \quad (3)$$

Here, $\hat{\epsilon}_p$ is a function of the RF driving frequency (ω_{RF}), the plasma oscillation frequency (ω_{pe}), the electron-neutral gas elastic collision frequency (ν_{en}), and the electron cyclotron frequency (ω_{ce}),^{1,9}

$$\hat{\epsilon}_p = \epsilon_0 \begin{bmatrix} \kappa_{\perp} & -j\kappa_{\times} & 0 \\ j\kappa_{\times} & \kappa_{\perp} & 0 \\ 0 & 0 & \kappa_{\parallel} \end{bmatrix}, \quad (4)$$

$$\begin{aligned} \kappa_{\perp} &= 1 - \frac{(\omega_{\text{RF}} - j\nu_{\text{en}})\omega_{\text{pe}}^2}{\omega_{\text{RF}}(\omega_{\text{RF}} - j\nu_{\text{en}})^2 - \omega_{\text{RF}}\omega_{\text{ce}}^2}, \\ \kappa_{\times} &= \frac{\omega_{\text{ce}}\omega_{\text{pe}}^2}{\omega_{\text{RF}}(\omega_{\text{RF}} - j\nu_{\text{en}})^2 - \omega_{\text{RF}}\omega_{\text{ce}}^2}, \\ \kappa_{\parallel} &= 1 - \frac{\omega_{\text{pe}}^2}{\omega_{\text{RF}}(\omega_{\text{RF}} - j\nu_{\text{en}})}. \end{aligned} \quad (5)$$

Here, ϵ_0 is the vacuum permittivity. $\kappa_{\perp}$, $\kappa_{\times}$, and $\kappa_{\parallel}$ are the perpendicular, off-diagonal, and parallel components of the tensor of the i th reaction. Because the ion oscillation frequency and the ion cyclotron frequency are much smaller than ω_{RF} and ω_{ce} , the motion of the ion is not considered in $\hat{\epsilon}_p$. The simulation calculates the components of $\vec{E}$ in radial (E_r), angular (E_{θ}), and axial (E_z) directions.³⁰ An inductive discharge can be calculated without B , and E_{θ} is significantly larger than E_r and E_z . However, because B changes the dielectric constant of plasma when it is applied, an

15 September 2025 21:32:22

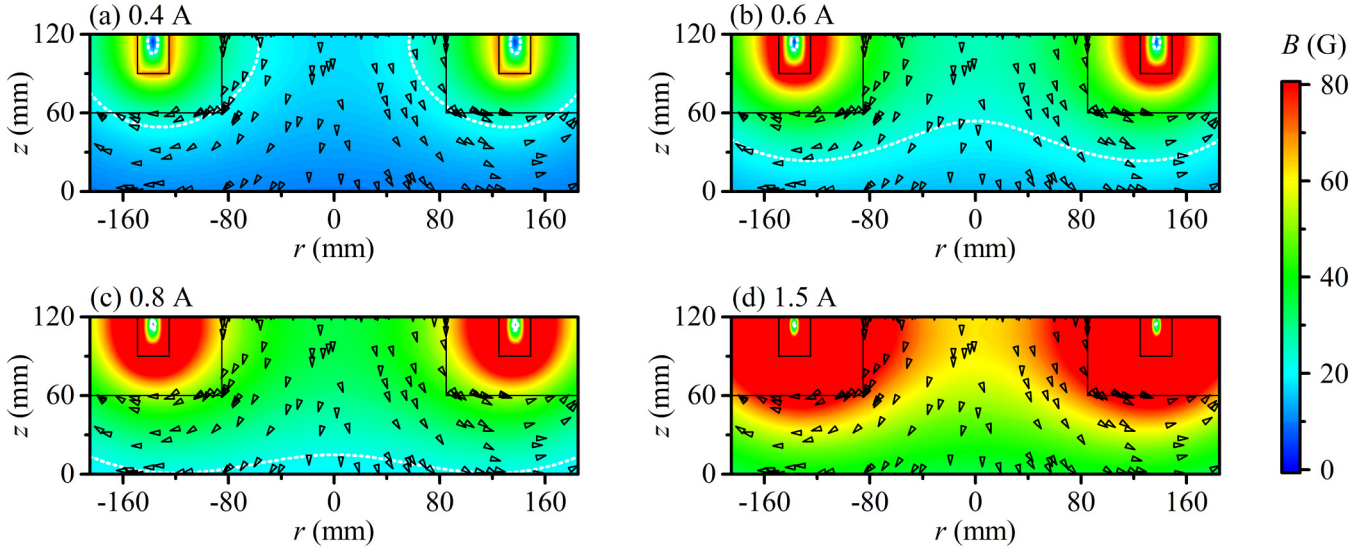


FIG. 3. Distributions of the magnetic field in the chamber for magnet coil currents of (a) 0.4, (b) 0.6, (c) 0.8, and (d) 1.5 A in the simulation. The color map, arrows, and black lines represent the magnitude, direction, and chamber edge, respectively. The white dot line represents the ECR magnetic condition 21.4 G.

amplitude for E_r similar to that of E_θ can be obtained from the simulation. These values of E_r and E_θ can be regarded as components of a right-hand polarized (RHP) wave, which can propagate in magnetized plasma at RF frequencies.^{1,2,9,31}

In the simulation, electron density is calculated from the continuity equation, as follows:

$$\frac{\partial n_e}{\partial t} + \nabla \cdot [-n_e(\hat{\mu}_e \cdot \vec{E}) - \hat{D}_e \cdot \nabla n_e] = \sum K_i n_{g,i} n_e, \quad (6)$$

$$\hat{\mu}_e^{-1} = \begin{bmatrix} \frac{1}{\mu_{dc}} & 0 & B_z \\ 0 & \frac{1}{\mu_{dc}} & -B_r \\ -B_z & B_r & \frac{1}{\mu_{dc}} \end{bmatrix}. \quad (7)$$

Because B can influence the transport of charged particles and can cause an anisotropy of electron mobility, the tensor of electron mobility ($\hat{\mu}_e$) is used, and $\hat{\mu}_e^{-1}$ is calculated using the radial and axial components of B (B_r and B_z) and the electron mobility in the DC electric field without B (μ_{dc}). μ_{dc} is calculated from $e/(m_e v_{en})$. $\hat{D}_e$, calculated as $\hat{\mu}_e T_e$, is the electron diffusion tensor. The right side of the equation is the sum of all the reactions of the electron generation, including the ionizations of the ground argon atoms and the metastable argon atoms; K_i and $n_{g,i}$ are the rate constant and gas density of the i th reaction, respectively. Substituting ion density (n_i), the tensors of ion mobility and ion diffusion constant ($\hat{\mu}_i$ and $\hat{D}_i$) with n_e ($\hat{\mu}_e$ and $\hat{D}_e$), the continuity equation for the ion can be obtained. Similar to $\hat{\mu}_e$, $\hat{\mu}_i$ is calculated from ion mass, ion temperature, ion-neutral gas collision frequency, and magnetic field. The electron energy distribution is assumed to be Maxwellian

distribution. Electron temperature is calculated from the energy conservation equation³²

$$\frac{\partial}{\partial t} \left(\frac{3}{2} n_e e T_e \right) + \nabla \cdot \left(\frac{5}{2} e T_e \Gamma_e - \chi \nabla T_e \right) = p_{abs} - \sum K_j n_{g,j} n_e \epsilon_j. \quad (8)$$

Here, Γ_e is the electron flux and χ is the electron thermal conductivity.^{9,32} In cold plasma, the electrons gain energy from the RF electric field and consume energy through collisions and by maintaining the electron temperature. Therefore, the first term on the right side of the above equation is the total power absorbed by the plasma per unit volume (p_{abs}), which is the sum of ohmic heating and ECR heating per unit volume ($p_{abs} = p_{ohm} + p_{ECR}$). The electrons consume the energy in elastic collision, excitation reaction, and ionization reaction, and their sum is represented by the second term on the right side of the above equation. The cross sections³³ of the elastic collision, excitation reaction, and ionization reaction, as well as the reactions of the metastable argon atom, are considered in the simulation. The gas temperature and ion temperature are considered as the constant of 300 K. The mesh used in the plasma region consists of 6055 nonuniform elements and 252 boundary elements.

At a pressure of 5 mTorr, ECR heating is the dominant term in the total power absorbed. ECR heating is calculated from the force equation of a single electron in the electrical field (E_{ECR})^{9,34}

$$\begin{aligned} \frac{dv_x}{dt} &= -\frac{eB}{m_e} v_y - \frac{eE_{ECR}}{m_e} \cos(\omega_{RF} t) \\ \frac{dv_y}{dt} &= \frac{eB}{m_e} v_x - \frac{eE_{ECR}}{m_e} \sin(\omega_{RF} t). \end{aligned} \quad (9)$$

The terms v_x and v_y are the electron velocity components in the x - and y directions, respectively. The solutions for these equations as a function of time, t , are as follows:

$$\begin{aligned} v_x(t) &= \frac{eE_{\text{ECR}}\sin(\omega_{\text{RF}}t)}{m_e(\omega_{\text{ce}} - \omega_{\text{RF}})} - c_2\sin(\omega_{\text{ce}}t) + c_1\cos(\omega_{\text{ce}}t) \\ v_y(t) &= -\frac{eE_{\text{ECR}}\cos(\omega_{\text{RF}}t)}{m_e(\omega_{\text{ce}} - \omega_{\text{RF}})} + c_1\sin(\omega_{\text{ce}}t) + c_2\cos(\omega_{\text{ce}}t) \\ c_1 &= v_x(0) \\ c_2 &= v_y(0) + \frac{eE_{\text{ECR}}}{m_e(\omega_{\text{ce}} - \omega_{\text{RF}})}. \end{aligned} \quad (10)$$

The constants c_1 and c_2 depend on $v_x(0)$ and $v_y(0)$, which are the initial values of v_x and v_y , respectively. The variation in electron kinetics ($\Delta\epsilon$) of a single electron at t is a function of t , $v_x(0)$, and $v_y(0)$,

$$\Delta\epsilon(t, v_x(0), v_y(0)) = \frac{m_e}{2} [v_x^2(t) + v_y^2(t) - v_x^2(0) - v_y^2(0)]. \quad (11)$$

The instantaneous power of the single electron can be defined by $d(\Delta\epsilon)/dt$. The total instantaneous power ($p_{\text{ECR}}(t)$) is a function of time. Considering the three-dimensional electron velocity distribution function ($g(v)$), $p_{\text{ECR}}(t)$ can be obtained by integrating $d(\Delta\epsilon)/dt$,

$$\begin{aligned} g(v) &= n_e \left(\frac{m_e}{2\pi kT_e} \right)^{3/2} \exp \left(-\frac{m_e(v_x^2 + v_y^2 + v_z^2)}{2kT_e} \right) \\ p_{\text{ECR}}(t) &= \iiint \frac{d}{dt} \Delta\epsilon(t, v_x(0), v_y(0)) g(v) dv_x dv_y dv_z \\ &= \iiint \left(-\frac{e^2 E^2}{m_e(\omega_{\text{ce}} - \omega_{\text{RF}})} \sin(\omega_{\text{ce}}t - \omega_{\text{RF}}t) - eEv_x(0)\cos(\omega_{\text{ce}}t - \omega_{\text{RF}}t) \right) g(v) dv_x dv_y dv_z \\ &= \frac{e^2 E^2 n_e}{m_e(\omega_{\text{ce}} - \omega_{\text{RF}})} \sin(\omega_{\text{ce}}t - \omega_{\text{RF}}t). \end{aligned} \quad (12)$$

Given that this heating process is disrupted by the collision with neutral gas,³⁴ during the average time interval between two collisions ($\tau = 1/\nu_{\text{en}}$), the time-averaged power density of ECR heating (p_{ECR}) can be derived from the ratio between the energy gained by ECR heating ($W_{\text{ECR}}(\tau)$) and τ ,

$$\begin{aligned} p_{\text{ECR}} &= \frac{1}{\tau} \int_0^\tau \frac{e^2 E^2 n_e}{m_e(\omega_{\text{ce}} - \omega_{\text{RF}})} \sin(\omega_{\text{ce}}t - \omega_{\text{RF}}t) dt \\ &= \frac{e^2 E^2 n_e \nu_{\text{en}}}{m_e(\omega_{\text{ce}} - \omega_{\text{RF}})} \left[-\frac{\cos(\omega_{\text{ce}}t - \omega_{\text{RF}}t)}{(\omega_{\text{ce}} - \omega_{\text{RF}})} \right]_0^\tau \\ &= \frac{e^2 E^2 n_e \nu_{\text{en}}}{m_e(\omega_{\text{ce}} - \omega_{\text{RF}})^2} \left(1 - \cos \left(\frac{\omega_{\text{ce}} - \omega_{\text{RF}}}{\nu_{\text{en}}} \right) \right). \end{aligned} \quad (13)$$

From this equation, a maximum value can be expected due to the ECR effect under the same E_{ECR} when the difference between ω_{ce} and ω_{RF} approaches 0.³ At higher pressure, a larger ν_{en} causes

the coefficient in the brackets of the p_{ECR} equation to approach 0. The lower p_{ECR} at a larger ν_{en} indicates that electron-neutral collisions reduce the heating effect of ECR.³ Because the cyclotron motion of electrons is perpendicular to B , the component of $\vec{E}$ that is perpendicular to B is used as E_{ECR} in plasma. Equation (13) is the power of ECR heating per unit volume, and at each position, E_{ECR} , n_e , ν_{en} , and ω_{ce} were used to calculate the p_{ECR} . Then, the sum of p_{ohm} and p_{ECR} was regarded as p_{abs} , which was taken into account of electron energy conservation equation [Eq. (8)] to obtain the electron temperature. The total RF power absorbed by plasma is defined as the volume integral of p_{abs} and was set as constant for all conditions of I_{mag} .

Because fluid modeling was used, the reliability of the fluid approximation should be discussed in the simulation setup of this article. The Knudsen number³⁵ ($K_n = \lambda_{\text{en}}/L$) can be used to evaluate the reliability of the fluid approximation, which is defined as the ratio of the mean free path of the electron-neutral collision (λ_{en}) to the characteristic length of the system (L). Because B was applied in the plasma, the electron mobility can be reduced by B and, the electron cyclotron radius (r_{ce}) can replace λ_{en} when r_{ce} is smaller than λ_{en} . In the radial direction, at 21 G of B_z and 5 V of T_e , r_{ce} was about 0.4 cm and the radius of chamber top part was 8.5 cm, K_n was 0.047. In the axial direction, the chamber gap length was 6 cm, and K_n was 0.067. These indicate that the simulation was in a near-continuous flow state, suggesting that the fluid model can still be used but with caution. The fluid model used can lead to the difference on T_e between simulation and measurement.

IV. RESULTS AND DISCUSSION

A. Distribution of electron temperature on a vertical section

When the RF power of the antenna was maintained at 300 W, the distributions of T_e on the vertical section of the chamber versus I_{mag} are shown in Figs. 4(a)–4(k). Overall, T_e increased with I_{mag} , with higher T_e below the antenna in the chamber top part and diffused along the magnetic field line into the chamber downstream part. This was mainly due to the enhanced heating of electrons by the RHP waves and the reduced diffusion cross B . Specifically, the variation of T_e can be divided into three stages. When I_{mag} increased from 0.4 to 0.9 A, a smaller increase of T_e was observed only in the chamber top part, and the confinement effect of electrons gradually enhanced. As I_{mag} increased from 1 to 1.2 A, an obvious increase in T_e was measured from the antenna along the magnetic field line, and T_e at the edge of the chamber downstream part also increased. At I_{mag} larger than 1.2 A, T_e increased in the entire chamber, particularly at the central axis ($r=0$ mm), and r was larger than 100 mm in the chamber downstream part. Similar variations of T_e were observed in the simulation. As shown in Fig. 5, the electrons in the simulation also gained energy at the chamber top part below the antenna and diffused along the magnetic field lines. There was a slight increase in T_e as I_{mag} increased from 0.6 to 0.9 A, an obvious increase in T_e when I_{mag} increased from 0.9 to 1.2 A, and a higher T_e in the entire chamber at 1.5 A.

In the weakly magnetized plasma, because B can create possibilities for wave propagation and the electron gains energy from $\vec{E}$, the variation of the intensity and distribution of $\vec{E}$ in the chamber

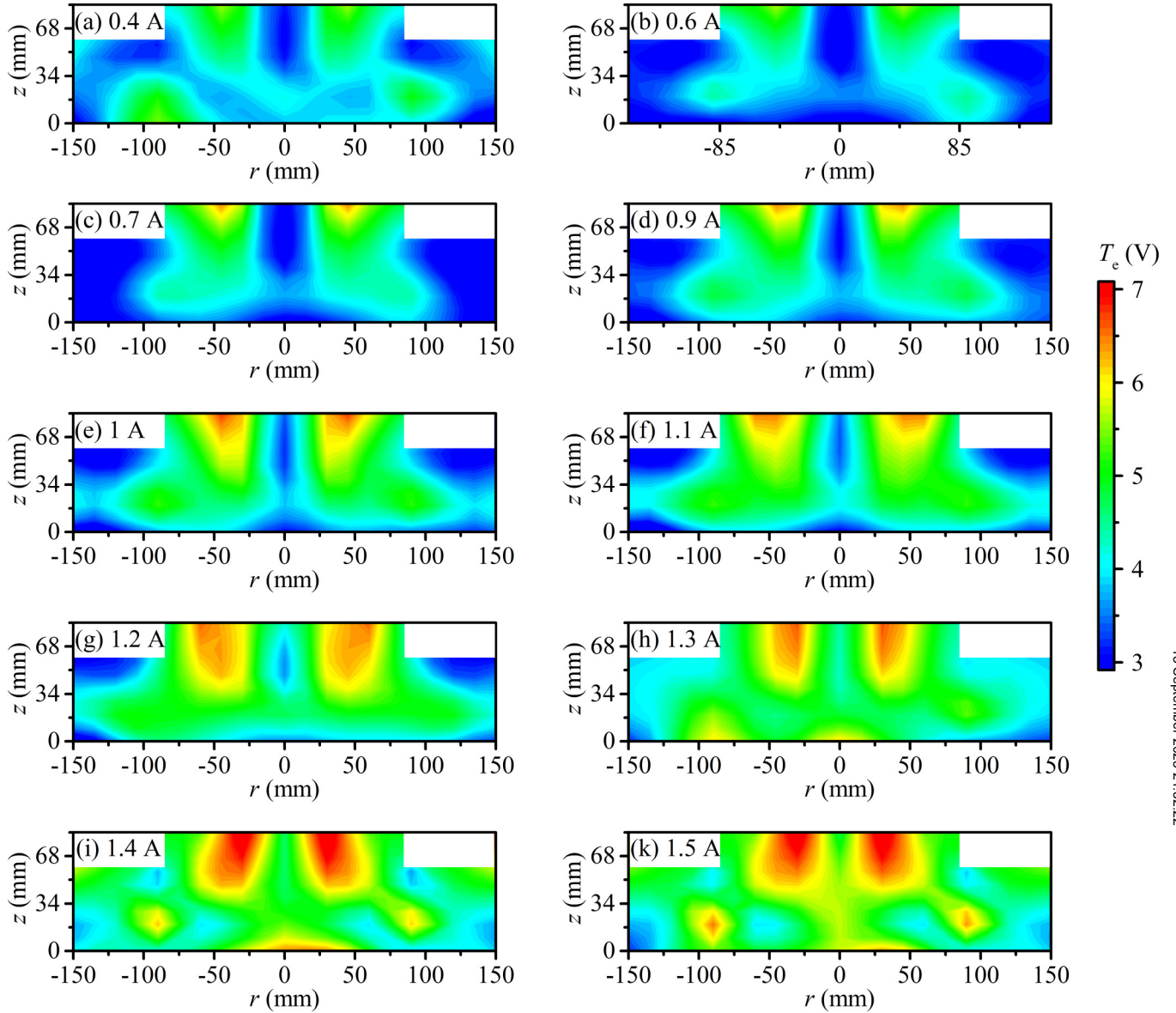


FIG. 4. Distributions of electron temperature measured for magnet coil currents of (a) 0.4, (b) 0.6, (c) 0.7, (d) 0.9, (e) 1, (f) 1.1, (g) 1.2, (h) 1.3, (i) 1.4, and (k) 1.5 A.

is a crucial factor in the variation of T_e . When the difference in n_e is not significant, the distribution of $\vec{E}$ depends on B in the weakly magnetized ICP. From the dispersion relation of the RHP wave in magnetized plasma, when the frequencies are much smaller than ω_{pe} , the RHP wave can propagate in the plasma when ω_{RF} is smaller than ω_{ce} .^{1,2} In Fig. 3, the position of the ECR magnetic condition shifted from the chamber top part to the bias electrode when I_{mag} increased from 0.4 to 0.9 A. In the range from the top quartz window to the bias electrode, B was larger than the ECR magnetic condition when I_{mag} was larger than 0.9 A. Therefore, when I_{mag}

was between 0.4 and 0.9 A, $\vec{E}$ could be cut off at the position where B was smaller than 21.4 G. As I_{mag} increased, electron heating slightly enhanced because the RHP waves penetrated deeper into the bulk plasma along the negative axial direction. When I_{mag} increased beyond 0.9 A, the RHP waves propagated from the antenna to the bias electrode and were reflected by the bias electrode. Thus, the standing wave can be expected to appear in the range between the top quartz window and the bias electrode.³⁶ In the range of I_{mag} from 0.9 to 1.2 A, the amplitude of $\vec{E}$ increased dramatically with I_{mag} , and electron heating enhanced significantly.

15 September 2025 21:32:22

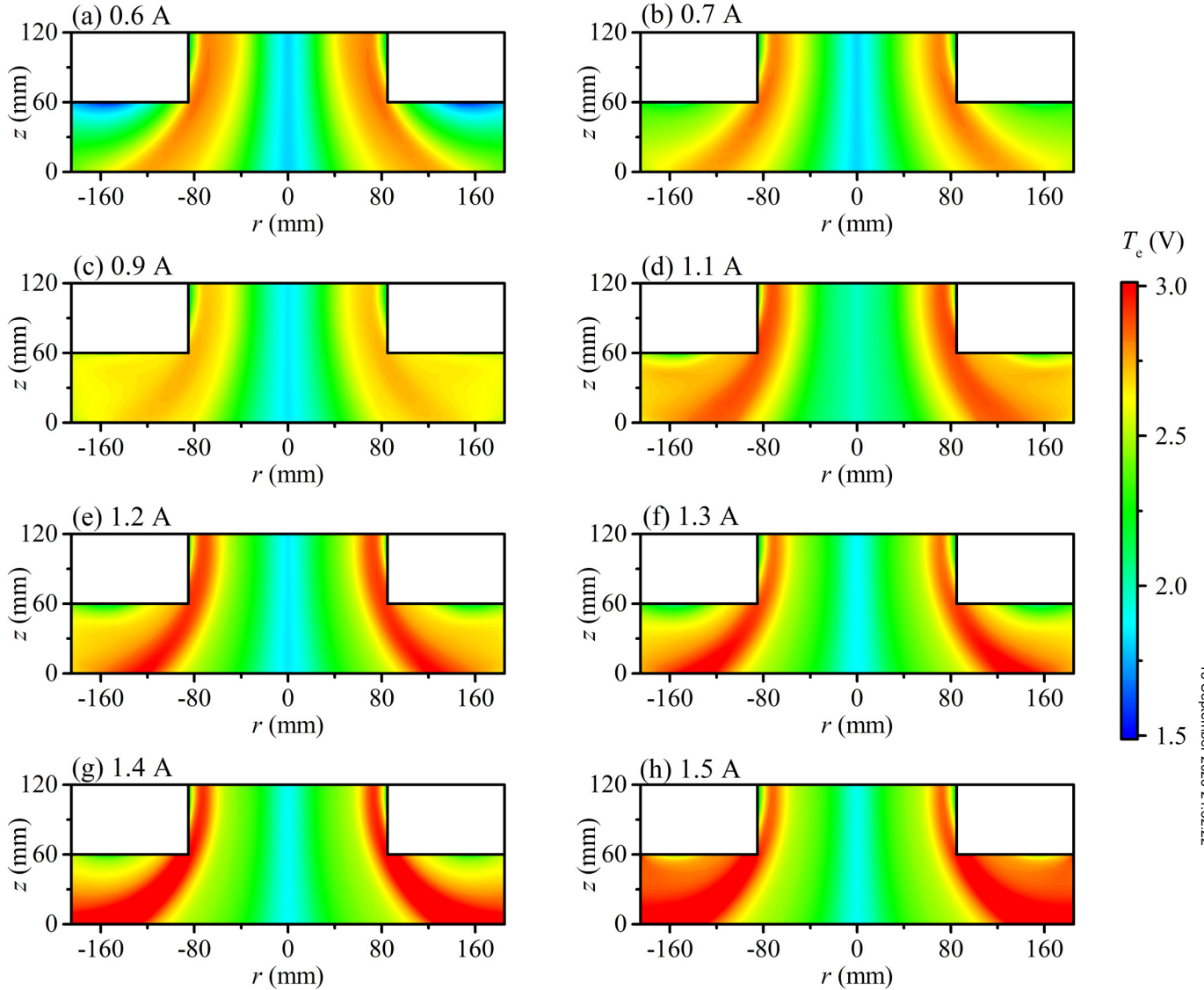
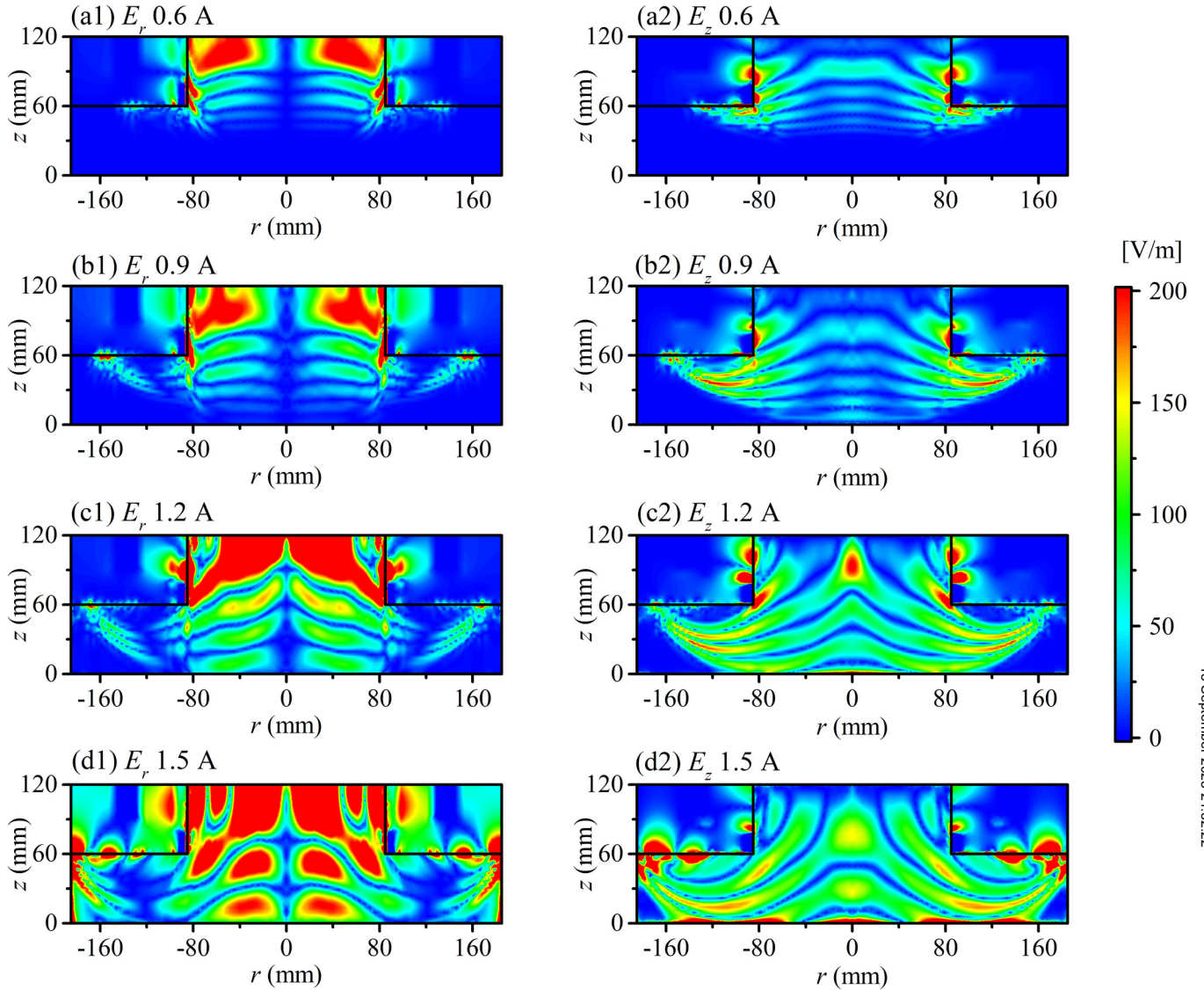


FIG. 5. Distributions of electron temperature in simulation for magnet coil currents of (a) 0.6, (b) 0.7, (c) 0.9, (d) 1.1, (e) 1.2, (f) 1.3, (g) 1.4, and (h) 1.5 A.

These variations on $\vec{E}$ can be observed in the simulation, and Fig. 6 shows amplitudes of E_r and E_z at 0.6, 0.9, 1.2, and 1.5 A of I_{mag} . Because the electric field components of the RHP waves in plasma were perpendicular to B , the main components of the RHP waves were E_r and E_θ at the chamber top part and the center of the chamber downstream part (-80 to 80 mm of r). At the edge of the chamber downstream part, E_z and E_θ were the main components of the RHP waves since the location was below the magnet coil. The standing wave effect of the RHP waves was observed in the region where B was larger than 21.4 G. For I_{mag} larger than 1.2 A, B was larger than 21.4 G in almost the entire chamber, and a stronger $\vec{E}$ at the edge of the chamber downstream part led to an increase in

T_e there. Particularly, around the central axis, there was an enhancement of E_z , and this enhanced E_z resulted in a higher T_e at the chamber central axis when I_{mag} was 1.9 A. E_z also showed a standing wave and was in the same phase as the RHP waves. Considering that the waves propagated from the top quartz window to the bias electrode, E_z was an electrostatic wave in plasma. Additional calculations of electromagnetic waves were performed considering the plasma as a uniform material within the chamber. Values of $B = 40$ G and $n_e = 2 \times 10^{11} \text{ cm}^{-3}$ were used in $\hat{\epsilon}_p$. An electrostatic wave on the chamber central axis was generated due to the geometry of the planar antenna, and the amplitude of E_z decreased with ω_{RF} .

15 September 2025 21:32:22



15 September 2025 21:32:22

FIG. 6. Distributions of the RF electric field amplitude in simulation for different magnet coil currents. (a1), (b1), (c1), and (d1) are the electric field radial components (E_r) for 0.6, 0.9, 1.2, and 1.5 A, respectively. (a2), (b2), (c2), and (d2) are the electric field axial components (E_z) for 0.6, 0.9, 1.2, and 1.5 A, respectively.

Comparing Figs. 4 and 5, it was found that T_e in the simulation was smaller than that in the experiment. First, at the ECR condition, the effect of ECR heating in the simulation was limited and weaker than that in the experiment. To prevent the value of $(\omega_{ce} - \omega_{RF})^2$ in the p_{ECR} equation from reaching 0, the minimum value of the difference between ω_{ce} and ω_{RF} was set to 10^6 because it was about 3.77×10^8 . Because a smaller difference between ω_{ce} and ω_{RF} resulted in larger p_{ECR} , the effect of ECR heating reduced at the ECR condition in the calculations. However, this limitation was not present in the experiment. Second, the simulation of magnetized ICP was performed in a 2D model, the powered and grounded nodes of the antenna could not be represented, and the

capacitive coupling between the antenna and the ground wall also could not be considered. In previous research,³⁷ the effect of capacitive coupling on the RF power absorbed by plasma and the electron density was much smaller than that of inductive coupling at 10^{11} cm^{-3} of electron density, but stochastic heating at the sheath-plasma interface could efficiently increase T_e around the chamber edge, such as the bias electrode. Stochastic heating was proportional to the square of ω_{RF} ,^{1,38,39} as the sheath-plasma surface could oscillate faster at the higher ω_{RF} . At ω_{RF} 60 MHz and I_{mag} 1.5 A, the difference in T_e between the simulation and the experiment around the bias electrode was significant. Third, different from the measurement, the change on T_e along the line of B was small in

simulation. The Lorentz force was perpendicular to the lines of B , and λ_{en} should be used when the Knudsen number was calculated. As mentioned in the simulation setup, in fluid modeling, the longer λ_{en} at 5 mTorr of pressure resulted in an inaccurate fluid approximate and the difference on T_e between simulation and measurement.

B. Distribution of ion density on a vertical section

The distributions of n_i on the vertical section in the chamber versus I_{mag} are shown by the contour map with the boundary

condition of $n_i = 0 \text{ cm}^{-3}$ in Figs. 7(a)–7(k). In all the conditions of I_{mag} , ions were generated at the chamber top part and diffused along the magnetic field lines to the chamber downstream part. Because the magnitude of B increased with I_{mag} , n_i changed in a more concentrated shape at the position below the antenna in the chamber top part, and the difference in n_i between the positions of $r = 0$ and below the antenna was more obvious. As for the n_i radial distribution at different values of z , the radial distribution of n_i gradually changed from a saddle shape to a platform shape when the plasma diffused from the top part to the downstream part in the chamber because of the weaker B in the chamber downstream

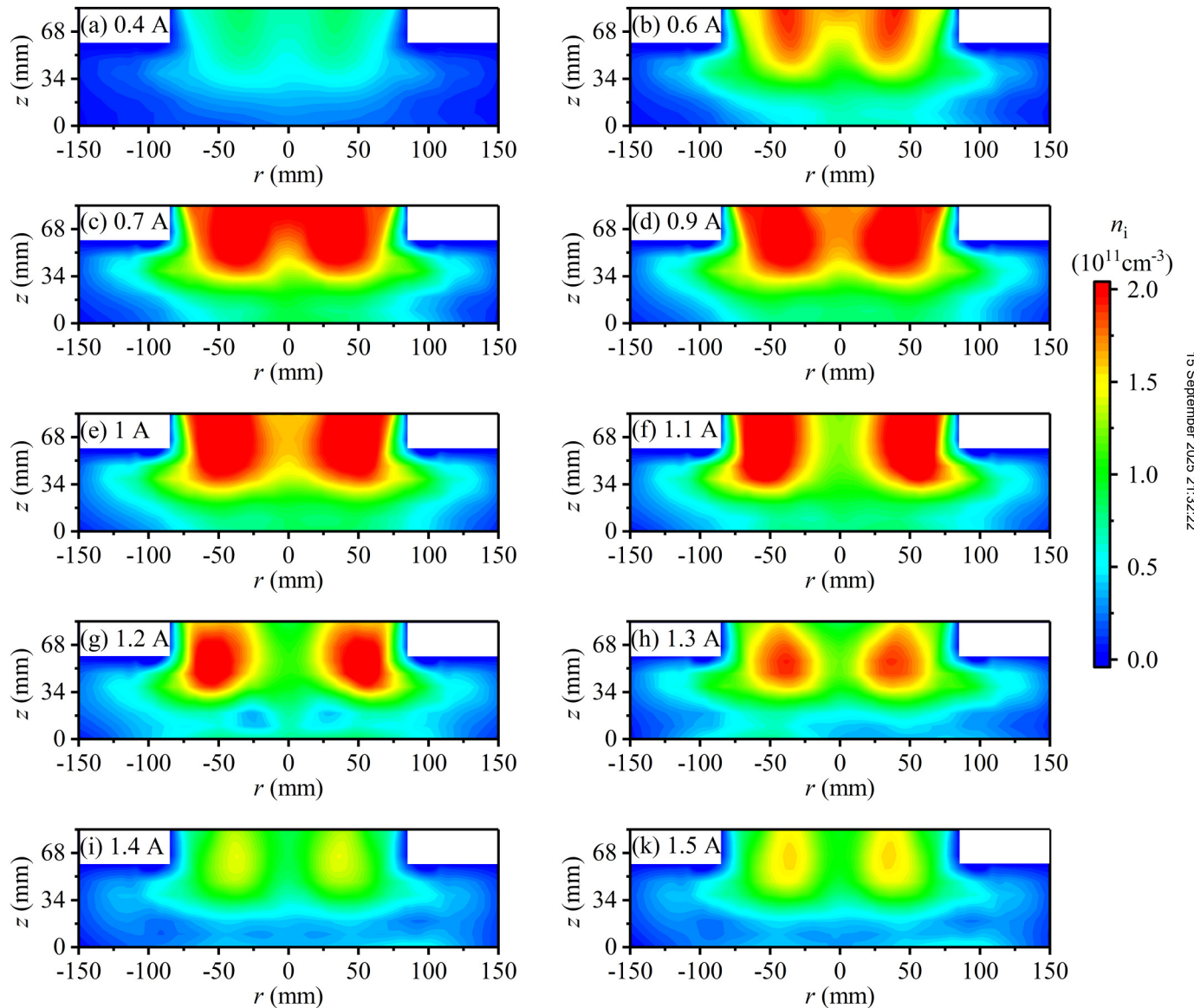


FIG. 7. Distributions of ion density measured for magnet coil currents of (a) 0.4, (b) 0.6, (c) 0.7, (d) 0.9, (e) 1, (f) 1.1, (g) 1.2, (h) 1.3, (i) 1.4, and (k) 1.5 A.

15 September 2025 21:32:22

part. However, at larger I_{mag} , an M-shaped n_i distribution could also be observed in the chamber downstream part. At z values of 85 and 45 mm, the radial distributions of n_i measured by the two probes exhibited the saddle shape for all conditions of I_{mag} . But at $z = 22$ mm, the radial distribution of n_i had a central-high shape in the range of I_{mag} from 0.4 to 0.7 A, the platform profile for I_{mag} values of 0.9–1.1 A, and the saddle shape when I_{mag} was larger than 1.1 A. On the bias electrode, the saddle shape n_i distribution was measured when I_{mag} was larger than 1.4 A. Therefore, the profile of n_i around the bias electrode could be controlled by I_{mag} . With the increase in I_{mag} , the maximum value of n_i increased first and then decreased at larger I_{mag} . The highest n_i was measured at

1 A of I_{mag} . Similar changes in n_i were observed in the simulation, as shown in Fig. 8. As I_{mag} increased, the n_i radial distribution changed to a more concentrated saddle shape, and the n_i maximum value also increased first and then decreased at larger I_{mag} . In the simulations, the peak value of n_i was at $I_{\text{mag}} = 0.9$ A, which was smaller than 1 A of I_{mag} in the experiment. As mentioned in Sec. II, B around the bias electrode in the experiment was smaller than that in the simulation when the arrangement of magnet coil wires was not considered. Therefore, B for the maximum n_i can be regarded as a similar condition in the experiment and the simulation, which is that B around the bias electrode is slightly larger than the ECR magnetic condition of 21.4 G.

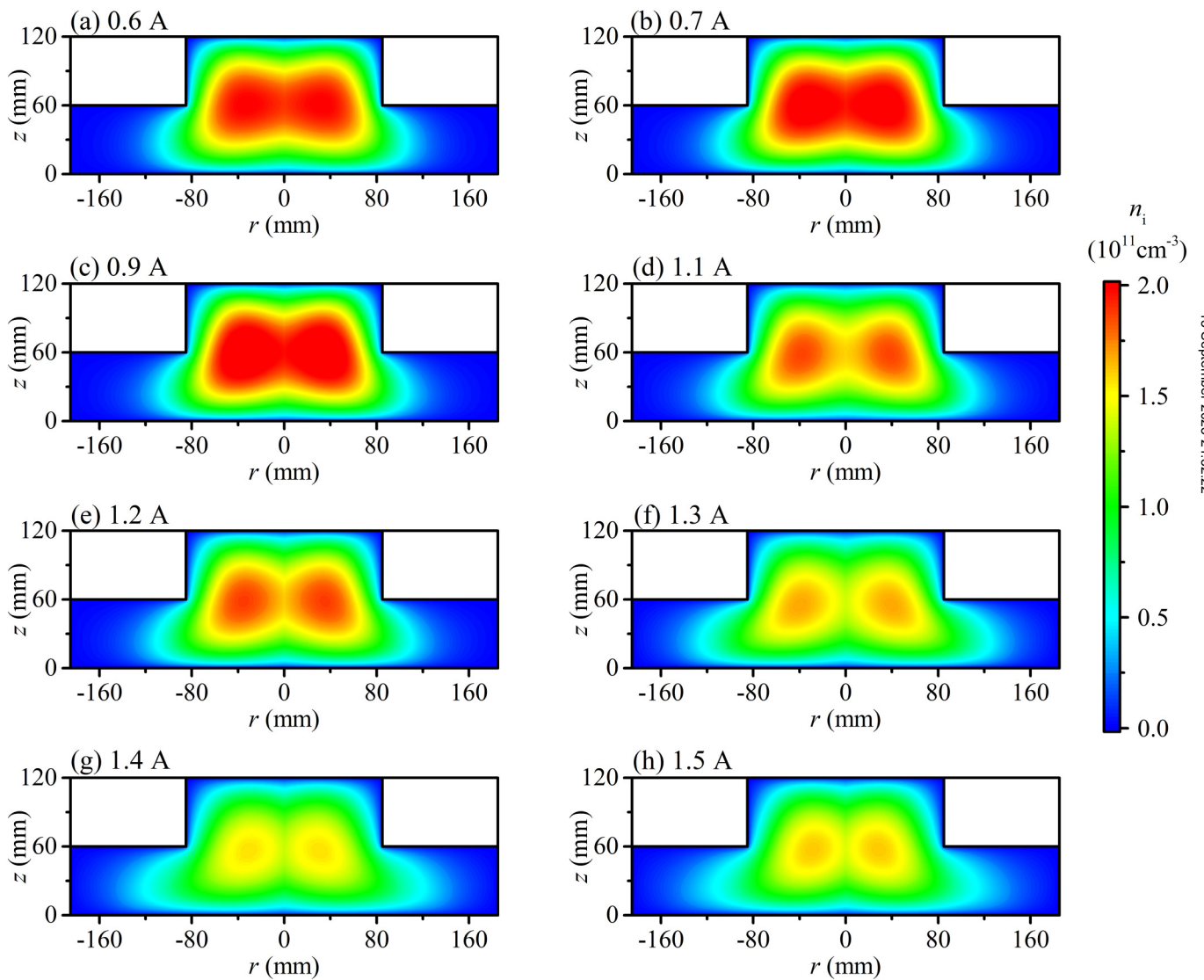
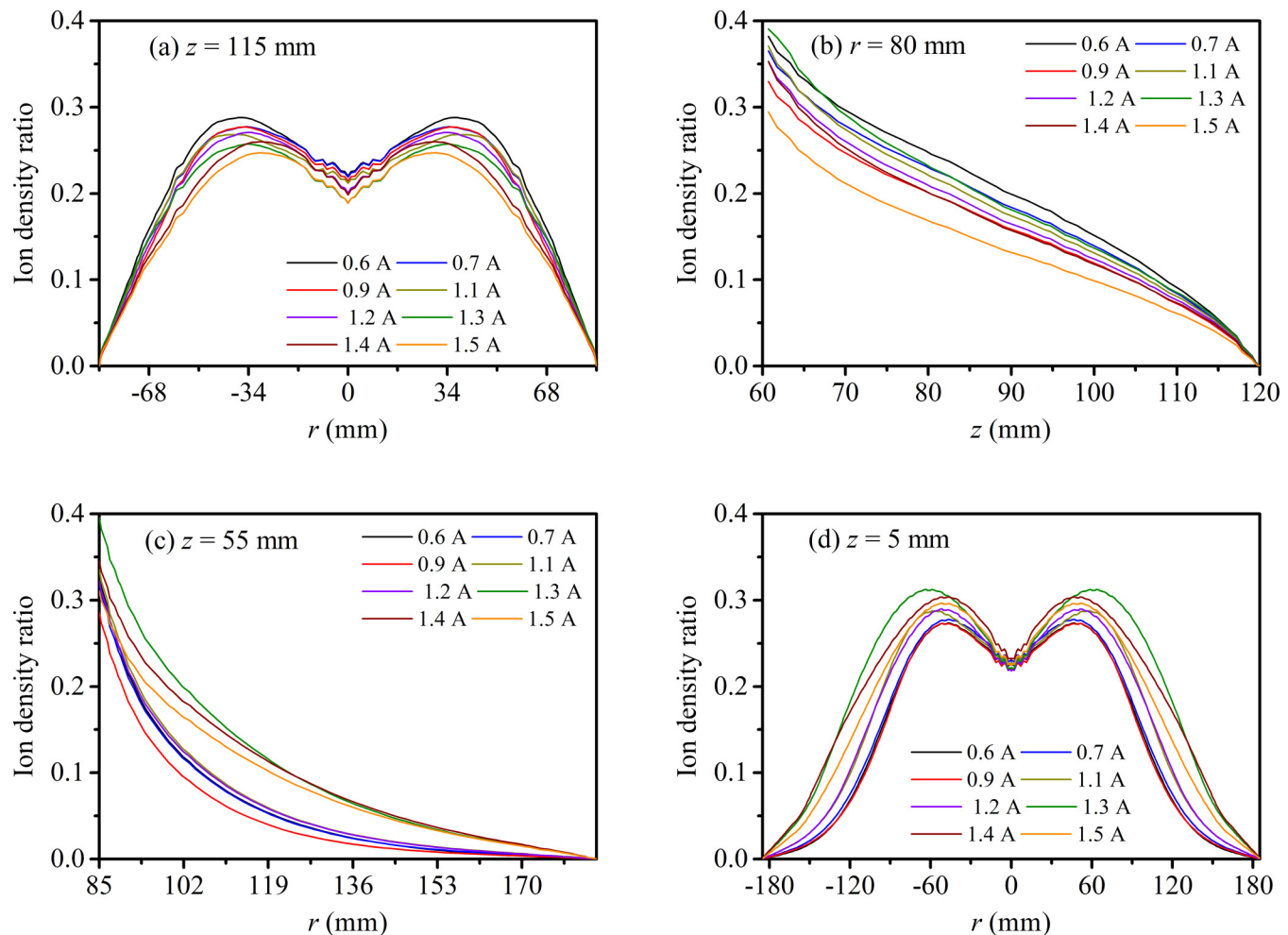


FIG. 8. Distributions of ion density in simulation for magnet coil currents of (a) 0.6, (b) 0.7, (c) 0.9, (d) 1.1, (e) 1.2, (f) 1.3, (g) 1.4, and (h) 1.5 A.

15 September 2025 21:32:22

When the RF power absorbed by plasma is constant, the plasma density depends on the effective area for particle loss (A_{eff}), u_B , and the total energy loss per electron-ion pair (ε_T).¹ As I_{mag} increased, the enhanced $\vec{E}$ resulted in the higher T_e , as shown in Figs. 4 and 5. This higher T_e led to larger u_B and higher plasma potential, resulting in the larger energy loss on the chamber wall and the lower n_i at larger I_{mag} . However, the increases in T_e , u_B , and ε_T cannot explain the variation of n_i . The main reason for the variation of the n_i maximum value was that the B influenced the plasma diffusion and changed the edge-to-center density ratio and A_{eff} . To investigate the effect of B on the plasma diffusion, the edge-to-center ion density ratio at different I_{mag} was calculated. In simulation at each I_{mag} , n_i at 5 mm from the chamber wall was selected as the edge ion density, and the maximum value of n_i was selected as the center ion density. The ion density ratios at the top quartz window of the chamber top part ($z = 115$ mm), the side wall

of the chamber top part ($r = 80$ mm), the top electrode of the chamber downstream part ($z = 55$ mm), and the bias electrode ($z = 5$ mm) are shown in Figs. 9(a)–9(d), respectively. Because the ion density ratio at the side wall of the chamber downstream part was at the level of 0.01 and the maximum value of that was less than 0.025, the particle loss at the side wall of the chamber downstream part was negligible. In the chamber top part, the ion density ratio at $z = 115$ mm and $r = 80$ mm decreased with I_{mag} . Due to the increase of I_{mag} , the gradient of B in the axial direction also increased, as shown in Figs. 2 and 3. The parallel force of the gradient $B^{\parallel}$ toward the negative axial direction on the charged particles also enhanced. Because the parallel force was proportional to the gradient of B and could push the charged particle into the region of the weaker B , the ion diffusion from bulk plasma to the top quartz window reduced, and the lower ion density ratio at larger I_{mag} can be observed in Fig. 9(a). For $r = 80$ mm, the increased I_{mag}



15 September 2025 21:32:22

FIG. 9. Edge-to-center ion density ratio at the different edges of the chamber: (a) the top quartz window of the chamber top part, (b) the side wall of the chamber top part, (c) the top electrode of the chamber downstream part, and (d) the bias electrode in the chamber downstream part.

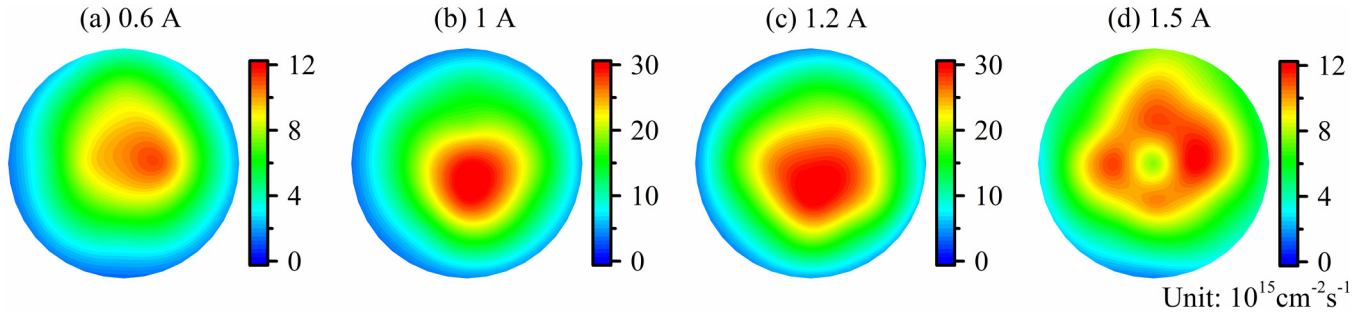


FIG. 10. Distributions of ion flux measured by the 2D probes on the bias electrode for magnet coil currents of (a) 0.6, (b) 1, (c) 1.2, and (d) 1.5 A.

and the larger B reduced the ion radial diffusion and the ion density ratio at $r = 80$ mm, as shown in Fig. 9(b). When I_{mag} increased, the parallel force effect and the confinement effect of B reduced the particle loss on the edge in the chamber top part; however, these were not the main reasons for the variations in the maximum n_e value. In the chamber downstream part, the lowest ion density ratio was observed at $I_{\text{mag}} = 0.9$ A at z values of 55 and 5 mm, shown as red lines in Figs. 9(c) and 9(d). Because the chamber downstream part was far from the antenna, n_i was maintained mainly by the ion diffusion from the chamber top part, which can be affected by the difference in ion density between the two parts. When B around the bias electrode was slightly larger than the ECR magnetic condition, p_{ECR} in the diffusion part reached a peak value, which also represents the peak value of the ion generation in the chamber downstream part relative to I_{mag} . E_{ECR} was weaker at smaller I_{mag} ; at larger I_{mag} , the difference between ω_{ce} and ω_{RF} was significant, although E_{ECR} was more intense. This enhanced ion generation in the chamber downstream part reduced the ion diffusion from the chamber top part and led to the lowest ion density ratio at the edge of the chamber downstream part. A_{eff} for the chamber was minimized when B around the bias electrode was slightly larger than the ECR magnetic condition, and n_i in the chamber presented the maximum value.

Figures 7 and 9 show that when I_{mag} increased from 1.2 to 1.9 A, n_i decreased and fluctuated rather than continuously decreased. An increase in n_i was observed in both experiment and simulation when I_{mag} increased from 1.4 to 1.5 A. B was larger than 21.4 G in the chamber discharge region when I_{mag} was larger than 1.2 A, and the RHP waves could propagate in the entire chamber. Therefore, n_i was also affected by the dispersion relation determined by the wavelength and chamber gap length. In some previous studies,^{40–42} the resonance of power deposition by the RHP waves could be achieved at low pressure when the chamber gap length was equal to an integer multiple of the RHP wavelength. n_i could increase due to this cavity resonance and could decrease in other conditions. During the increase of I_{mag} , the RHP wavelength became longer, as shown in Fig. 6. Therefore, n_i decreased and fluctuated as I_{mag} increased.

C. Distribution of ion flux on the electrode

The performance of plasma processing generally depends on the ion flux and the ion energy on the bias electrode^{43–48} because

the chemical reaction can be enhanced by the larger ion flux and higher ion energy. Because bias power was not used in this experiment, the ion energy could be regarded as the sheath potential on the floating wall,¹ which was proportional to T_e . The measured ion flux distributions for 0.6, 1, 1.2, and 1.5 A of I_{mag} are shown in Figs. 10(a)–10(d), respectively. According to the measurement of n_i and T_e , the maximum value of n_i was observed at $I_{\text{mag}} = 1$ A, and T_e increased with I_{mag} . The ion flux was proportional to n_i and the square root of T_e . Because T_e increased with I_{mag} , the maximum ion flux was observed at 1.2 A of I_{mag} . Therefore, the maximum ion flux on the bias electrode could be obtained when B was slightly larger than the magnetic condition of maximum n_i . As in Sec. IV B, the central-high profile distribution of ion flux was measured at lower I_{mag} , and a larger ion flux ring with a radius of about 50 mm could be observed at $I_{\text{mag}} = 1.5$ A due to the larger B around the bias electrode. The distribution of ion flux on the bias electrode could be controlled by the antenna profile at the larger B .

V. CONCLUSIONS

The distributions of T_e and n_i on the vertical section in the ICP with the divergent magnetic field were observed by the radially movable floating probes placed at different axial positions. With the increases of I_{mag} and the magnetic field, the varied RF electric field increased T_e , and at higher I_{mag} , the electrostatic waves at the chamber center axial could result in the higher T_e there. The changed ECR heating in the chamber diffusion part resulted in the maximized n_i when the magnetic field around the bias electrode was slightly larger than the ECR magnetic condition. As T_e increased with I_{mag} , the maximum ion flux on the bias electrode was achieved at I_{mag} larger than that of the maximum n_i , and the distribution of ion flux could be controlled by the profile of the antenna at the larger magnetic field.

ACKNOWLEDGMENTS

This work was supported by the National Research Foundation of Korea (NRF) Grant funded by the Korea Government (MSIT) (No. RS-2023-00260527), the Ministry of Trade, Industry & Energy (Nos. 20020392, 20022472, 1415187403, 20024017, 20012609, 20018979, 20018982, 00154752, and 1415187722), and the Korea Semiconductor Research Consortium

15 September 2025 21:32:22

(KSRC) (Nos. RS-2023-00237362 and RS-2023-00235950). Following are the results of a study on the “Leaders in INdustry-university Cooperation 3.0” Project, supported by the Ministry of Education and National Research Foundation of Korea.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

You He: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (equal); Writing – original draft (lead). **Yi-Lang Jiang:** Data curation (supporting); Investigation (supporting); Software (equal). **Myoung-Jae Lee:** Funding acquisition (equal); Supervision (equal). **Beom-Jun Seo:** Software (supporting). **Ju-Ho Kim:** Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Writing – review & editing (equal). **Chin-Wook Chung:** Conceptualization (equal); Data curation (equal); Funding acquisition (lead); Supervision (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

REFERENCES

- ¹M. A. Lieberman and A. J. Lichtenberg, *Principles of Plasma Discharges and Materials Processing*, 2nd ed. (Wiley-Interscience, Hoboken, NJ, 2005).
- ²P. Chabert and N. Braithwaite, *Physics of Radio-Frequency Plasmas* (Cambridge University, Cambridge, 2011).
- ³C. Chung, S. S. Kim, and H.-Y. Chang, *Phys. Rev. Lett.* **88**, 095002 (2002).
- ⁴F. F. Chen, *Plasma Sources Sci. Technol.* **24**, 014001 (2015).
- ⁵D. Kim, H. Lee, B. Park, H. Kim, Y. Sung, S. Chae, Y. Ko, and G. Yeom, *Thin Solid Films* **447**, 100 (2004).
- ⁶V. A. Godyak and B. M. Alexandrovich, *Phys. Plasmas* **11**, 3553 (2004).
- ⁷J. W. Kim, H. W. Cheong, Y. T. Hong, and K. W. Whang, *Plasma Sources Sci. Technol.* **26**, 035008 (2017).
- ⁸Y. J. Ji *et al.*, *Nanotechnology* **35**, 275701 (2024).
- ⁹T. V. Tsankov, P. Chabert, and U. Czarnetzki, *Plasma Sources Sci. Technol.* **31**, 084007 (2022).
- ¹⁰C. Chung, S. S. Kim, and H.-Y. Chang, *Phys. Rev. E* **69**, 016406 (2004).
- ¹¹H.-J. Lee, I.-D. Yang, and K.-W. Whang, *Plasma Sources Sci. Technol.* **5**, 383 (1996).
- ¹²M.-H. Lee and C.-W. Chung, *Appl. Phys. Lett.* **93**, 151503 (2008).
- ¹³Y. H. Lee, J. E. Heidenreich III, and G. Fortunato, *J. Vac. Sci. Technol. A* **7**, 903 (1989).
- ¹⁴N. Sadeghi, T. Nakano, D. J. Trevor, and R. A. Gottscho, *J. Appl. Phys.* **70**, 2552 (1991).

- ¹⁵M. G. Blain, G. D. Tipton, W. M. Holber, G. S. Selwyn, P. L. Westerfield, and K. L. Maxwell, *Plasma Sources Sci. Technol.* **3**, 325 (1994).
- ¹⁶A. Ganguli, R. D. Tarey, N. Arora, and R. Narayanan, *Plasma Sources Sci. Technol.* **25**, 025026 (2016).
- ¹⁷H. Tamura, T. Tetsuka, D. Kuwahara, and S. Shinohara, *IEEE Trans. Plasma Sci.* **48**, 3606 (2020).
- ¹⁸K. Racka-Szmidt, B. Stonio, J. Żelazko, M. Filipiak, and M. Sochacki, *Materials* **15**, 123 (2022).
- ¹⁹R. L. Kinder and M. J. Kushner, *J. Appl. Phys.* **90**, 3699 (2001).
- ²⁰K. Takahashi, C. Charles, and R. W. Boswell, *Phys. Rev. Lett.* **110**, 195003 (2013).
- ²¹Y. Zhang, C. Charles, and R. Boswell, *Phys. Rev. Lett.* **116**, 025001 (2016).
- ²²D. Tang and P. K. Chu, *J. Appl. Phys.* **93**, 5883 (2003).
- ²³D. L. Tang, R. K. Y. Fu, X. B. Tian, and P. K. Chu, *Rev. Sci. Instrum.* **74**, 2704 (2003).
- ²⁴C. Charles and R. W. Boswell, *IEEE Trans. Plasma Sci.* **36**, 2141 (2008).
- ²⁵N. Gulbrandsen and Å. Fredriksen, *AIP Conf. Proc.* **5**, 2 (2017).
- ²⁶S. Yadav, K. K. Barada, S. Ghosh, J. Ghosh, and P. K. Chattopadhyay, *Phys. Plasmas* **26**, 082109 (2019).
- ²⁷T. Furukawa, Y. Yarita, H. Aoyagi, and H. Nishida, *J. Appl. Phys.* **131**, 173302 (2022).
- ²⁸T.-W. Kim, M.-Y. Lee, Y.-H. Hong, M.-H. Lee, J.-H. Kim, and C.-W. Chung, *Plasma Sources Sci. Technol.* **30**, 025002 (2021).
- ²⁹M.-H. Lee, S.-H. Jang, and C.-W. Chung, *J. Appl. Phys.* **101**, 033305 (2007).
- ³⁰H.-J. Lee and Y.-G. Kim, *Thin Solid Films* **521**, 78 (2012).
- ³¹F. F. Chen, *Introduction to Plasma Physics and Controlled Fusion* (Springer, New York, 1984).
- ³²A. N. Kropotkin and D. G. Voloshin, *Phys. Plasmas* **27**, 053507 (2020).
- ³³A. V. Phelps and Z. L. Petrovic, *Plasma Sources Sci. Technol.* **8**, R21 (1999).
- ³⁴J.-R. Liu, Y.-X. Liu, and Y.-N. Wang, *Plasma Sources Sci. Technol.* **32**, 044004 (2023).
- ³⁵G. Karniadakis, A. Beskok, and N. Aluru, *Microflows and Nanoflows: Fundamentals and Simulation* (Springer Science & Business Media, New York, 2006).
- ³⁶S. S. Kim, C. S. Chang, N. S. Yoon, and K.-W. Whang, *Phys. Plasmas* **6**, 2926 (1999).
- ³⁷J.-H. Kim and C.-W. Chung, *Phys. Plasmas* **27**, 023503 (2020).
- ³⁸M. A. Lieberman, *IEEE Trans. Plasma Sci.* **16**, 638 (1988).
- ³⁹B. P. Wood, M. A. Lieberman, and A. J. Lichtenberg, *IEEE Trans. Plasma Sci.* **23**, 89 (1995).
- ⁴⁰M. A. Lieberman and R. W. Boswell, *J. Phys. IV* **8**, Pr7-145 (1998).
- ⁴¹K.-K. Chi, T. E. Sheridan, and R. W. Boswell, *Plasma Sources Sci. Technol.* **8**, 421 (1999).
- ⁴²Y. Celik, D. L. Crintea, D. Luggenhölscher, and U. Czarnetzki, *Plasma Phys. Contr. Fusion* **51**, 124040 (2009).
- ⁴³D. A. Carl, D. W. Hess, and M. A. Lieberman, *J. Appl. Phys.* **68**, 1859 (1990).
- ⁴⁴J. Hopwood, *Plasma Sources Sci. Technol.* **1**, 109 (1992).
- ⁴⁵J. H. Keller, J. C. Forster, and M. S. Barnes, *Journal of Vac. Sci. Technol. A* **11**, 2487 (1993).
- ⁴⁶K. Takechi and M. A. Lieberman, *J. Appl. Phys.* **90**, 3205 (2001).
- ⁴⁷M. Klick and H.-P. Maucher, *J. Vac. Sci. Technol. B* **40**, 012203 (2022).
- ⁴⁸Y. He, Y.-L. Jiang, J. Jung, M.-S. Kim, J.-H. Kim, and C.-W. Chung, *Plasma Sources Sci. Technol.* **32**, 075008 (2023).

15 September 2025 21:32:22